Assignment 3

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Batch A3

Question 1

$$\overbrace{a+b+c}^{\text{First Group}} + \underbrace{x+y+z}_{\text{Second Group}} = \overbrace{a+\underbrace{b+c}_{\text{Sum 1}}}^{\text{Total}}$$

Question 2

$$A = \left[\begin{array}{c|ccc} & x_1 & x_2 & x_3 \\ \hline y_1 & a_{11} & a_{12} & a_{13} \\ y_2 & a_{21} & a_{22} & a_{23} \\ y_3 & a_{31} & a_{32} & a_{33} \end{array}\right]$$

Question 3

$$f(x) = \begin{cases} x^2 + 2x + 1, & x < 0, \\ \sqrt{x} + 1, & 0 \le x < 4, \\ \dfrac{x^2 - 1}{x - 1}, & x \ge 4. \end{cases}$$

Question 4

$$\left[\begin{array}{ccccc} \overbrace{\phantom{a_{11}\ a_{12}\ a_{13}}}^{\text{Column Group 1}} & & & \overbrace{\phantom{a_{14}\ a_{15}}}^{\text{Column Group 2}} & \\ a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \end{array}\right]$$

Question 5

$$F(x) = \underbrace{\sum_{i=1}^{n}(a_i x_i^2 + b_i x_i + c_i)}_{\text{Quadratic Contribution}} + \underbrace{\begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} A \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_{\text{Matrix Contribution}}$$

$$\underbrace{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad}_{\text{Total Energy}}$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}.$$

Question 6

Roll No.	**Name**	**Department**	**Semester**	**CGPA**
101	Amit	CSE	2	8.45
102	Neha	CSE	2	9.12
103	Rahul	ECE	4	8.76
104	Priya	ECE	4	9.01

Question 7

Academic Performance		
Student	Midterm	Final
A	78	89
B	85	92
C	91	94

Question 8

Department	Course	Credits	Grade
CSE	Data Structures	4	A
	Machine Learning	4	A+
	Operating Systems	3	B+
ECE	Digital Electronics	4	A
	Signals	3	B+

Question 9

Model	Parameters	FLOPs	Accuracy	Energy (kWh)
ResNet18	11.7M	1.8G	94.2%	0.00093
MobileNetV2	3.5M	0.3G	92.8%	0.00061
EfficientNet-B0	5.3M	0.39G	93.5%	0.00072

Question 10

Student	Mathematics	Physics	Computer Science
Amit	92	81	95
Neha	72	88	91
Rahul	68	74	79
Priya	96	94	98

Question 11

Technique	Description	Advantages	Limitations
Pruning	Removes unnecessary parameters from a neural network.	Reduces model size and computation.	May reduce accuracy.
Quantization	Represents weights using lower precision.	Reduces memory and energy consumption.	Can introduce numerical error.
Knowledge Distillation	Transfers knowledge from a large teacher model to a smaller student model.	Produces compact models.	Requires training a teacher model.

Question 12

Model	Parameters	FLOPs	Accuracy	Energy
ResNet18	11.7M	1.82G	94.21	0.00093
MobileNetV2	3.5M	0.30G	92.84	0.00061
EfficientNet-B0	5.3M	0.39G	93.52	0.00072

Table 1: Scientific model comparison

Question 13

Model	**Architecture**	**Performance**	
CNN	Conv $\rightarrow$ ReLU $\rightarrow$ Pool	Metric	Value
		Accuracy	94%
		F1 Score	93%
ResNet	Conv $\rightarrow$ BN $\rightarrow$ ReLU	Metric	Value
		Accuracy	96%
		F1 Score	95%
ViT	Attention $\rightarrow$ MLP	Metric	Value
		Accuracy	97%
		F1 Score	96%

Question 14

Model	**Loss Function**	**Parameters**	**Complexity**
Linear	$L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$	10K	$O(n)$
CNN	$L = -\sum y \log(\hat{y})$	1.2M	$O(n^2)$
Transformer	$L = -\sum_i y_i \log p_i$	110M	$O(n^2 d)$

— End of Assignment —